

Clinical Evaluation and Management of Chronic Kidney Disease

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Although many patients with chronic kidney disease (CKD) progress to end-stage kidney disease (ESKD) and require kidney replacement therapy (KRT), the majority die of nonrenal causes, particularly premature cardiovascular (CV) events.¹ Early diagnosis of CKD is therefore important because it provides opportunities to delay progression of CKD (see [Chapter 82](#)) and prevent CV complications (see [Chapters 84 and 85](#)).

DEFINITIONS

CKD is defined as abnormalities of kidney structure or function, present for at least 3 months, with implications for health ([Table 83.1](#)). The relevant Kidney Disease: Improving Global Outcomes (KDIGO) guideline recommends classification of CKD based on cause, category of glomerular filtration rate (GFR), and albuminuria (see [Fig. 82.2](#)).² Because of the impracticalities of using radioisotopes and 24-hour urine collections, the KDIGO classification system recommends that kidney function be assessed by estimating GFR (eGFR) from the serum creatinine concentration using an appropriate equation, except in circumstances in which eGFR estimations are known to be less accurate, such as when there is significant muscle wasting. Initially, the Modification of Diet in Renal Disease (MDRD) equation was used, but this has predominantly been replaced by the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation, which more accurately categorizes the risk for mortality and progression to ESKD (see [Chapter 3](#)).³ Recently, this formula has been updated in order to eliminate race from the equation (https://www.kidney.org/professionals/kdoqi/gfr_calculator/formula), and although the formula is less precise than the earlier CKD-EPI equation, it is sufficiently accurate for clinical practice. Although staging systems for CKD based on eGFR have limitations, they have proved useful in many clinical settings and are now deeply embedded in KDIGO and other guidelines developed for CKD management and in research.

The evidence base for the management of CKD is evolving. Although every effort has been made to ensure this chapter reflects current recommendations, the reader is advised to check for any new clinical trials and relevant guideline updates.

CLINICAL PRESENTATION

CKD is commonly detected by routine blood testing and is usually asymptomatic until late stage G4 or stage G5. Symptoms of CKD are nonspecific and need to be asked about directly ([Box 83.1](#)). There is some evidence that early diagnosis with appropriate management may slow the rate of decline of kidney function and reduce CV risk.⁴ Screening of the general population for CKD is not recommended, but in the UK, the National Institute for Health and Care Excellence (NICE) proposes offering testing to people with conditions associated

with an increased prevalence of CKD, which is termed *case-finding*. Subgroups of interest include those with diabetes, hypertension, previous acute kidney injury (AKI), CV disease (CVD), structural kidney tract disease, kidney calculi, prostatic hypertrophy, multisystem diseases with potential kidney involvement (e.g., systemic lupus erythematosus), a family history of category G5 CKD, and hereditary kidney disease, and after opportunistic detection of hematuria or proteinuria.⁵

Evaluation of Chronic Kidney Disease

Establishing Chronicity

When eGFR of less than 60 mL/min/1.73 m² is detected, careful attention should be paid to previous blood and urine test results and the clinical history to determine whether this is a result of AKI (i.e., an abrupt decrease in kidney function or CKD that has been present but asymptomatic for some time).

A detailed medical history covering issues, including other medical conditions, family history of kidney disease, prescribed medication, and recreational drug use, may suggest an underlying cause. There may be hints of a history of kidney problems (e.g., hypertension, proteinuria, microhematuria) or symptoms suggestive of prostatic disease. The findings of a physical examination are not always helpful, although skin pigmentation, scratch marks, left ventricular hypertrophy, and hypertensive fundal changes favor a chronic presentation ([Fig. 83.1](#)). Details of the social and personal circumstances are also crucial, particularly for patients with progressive kidney disease in whom KRT is likely to be required.

Blood tests for other conditions can be helpful because they may indicate evidence of an acute illness that could be the cause of kidney failure, such as systemic vasculitis or multiple myeloma. A normochromic normocytic anemia is usual in CKD but also may be a feature of acute systemic illnesses and therefore is not discriminatory. Low serum calcium and raised phosphate levels also have little discriminatory value, but normal levels of parathyroid hormone (PTH) are more in keeping with AKI. Patients with grossly abnormal biochemical values (e.g., blood urea > 300 mg/dL [>50 mmol/L] and/or serum creatinine > 13.5 mg/dL [>1200 μmol/L]) who appear relatively well and are still passing normal volumes of urine are much more likely to have CKD than AKI.

Assessment of Glomerular Filtration Rate

For patients in whom the distinction between AKI and CKD is unclear, repeat testing of kidney function should be performed within 2 weeks of the initial finding of eGFR less than 60 mL/min/1.73 m². However, if previous results confirm that this is a chronic finding, or if repeated blood test results over a 3-month period are consistent, CKD is confirmed. Other tests (such as cystatin C or an isotope-clearance measurement of GFR) can be used to confirm CKD in circumstances in which eGFR based on serum creatinine is known to be less accurate.

TABLE 83.1 Criteria for Definition of Chronic Kidney Disease (CKD)

CKD is defined as abnormalities of kidney structure or function, present for more than 3 months, with implications for health. These may include the following:

Markers of kidney damage	Albuminuria (AER $\geq$ 30 mg/24 h; uACR $\geq$ 30 mg/g [$\geq$ 3 mg/mmol])
	Urine sediment abnormalities
	Electrolyte and other abnormalities caused by tubular disorders
	Abnormalities detected through histology
	Structural abnormalities detected through imaging
	History of kidney transplantation
Decreased GFR	GFR $<$ 60 mL/min/1.73 m ²

AER, Albumin excretion rate; GFR, glomerular filtration rate; uACR, urinary albumin-to-creatinine ratio.

From Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.

BOX 83.1 Symptoms and Signs of Severe Chronic Kidney Disease

- Difficulty sleeping
- Nocturia
- Headache
- Restless leg syndrome
- Metallic taste in the mouth
- Shortness of breath on exertion or at rest, paroxysmal nocturnal dyspnea
- Fatigue, often profound
- Muscle cramps and twitches
- Seizures
- Lack or loss of appetite for food, abdominal pain, nausea, vomiting, and weight loss
- Itch, particularly on the trunk and worse at night
- Altered respiration including Kussmaul breathing and Cheyne-Stokes respiration
- Icteric sclera or “red eye” because of calcium deposition
- Muscle weakness
- Oral lesions including gingival bleeding or petechiae, xerostomia, periodontitis, and candidiasis
- Pericardial and/or pleural rub
- Pulmonary and peripheral edema
- Skin changes including xerosis (abnormal dryness), scratch marks, pallor, sallow coloration or hyperpigmentation
- Uremic flap (asterixis)
- Uremic fetor: Ammonia or urine-like odor to the breath
- Uremic frost: Crystallized urea deposits that can be found on the skin

Assessment of Proteinuria or Albuminuria

Dipstick testing of the urine and urine culture may reveal microhematuria, which can be a useful pointer toward an underlying diagnosis.⁶ Workup of hematuria is discussed in [Chapters 4 and 63](#). Whether or not proteinuria is detected by dipstick, there should be a further measurement of urinary albumin excretion. Albuminuria is an important diagnostic and prognostic marker, and its presence indicates a higher risk for both progression of kidney disease and CV complications.⁷ KDIGO guidelines recommend that the preferred method of assessing proteinuria is by measurement of the urinary albumin-to-creatinine ratio (uACR) using an early morning urine sample where practical.²



Fig. 83.1 Uremic Pigmentation. Diffuse brown pigmentation as seen here suggests chronic kidney disease rather than acute kidney injury.

The degree of albuminuria is graded by the A1 to A3 category system, replacing previous terms such as *microalbuminuria* (see [Fig. 82.1](#)). However, it is important to be aware that some patients will excrete proteins other than albumin, and a urine protein-to-creatinine ratio (uPCR) may be more useful for certain conditions.⁸ Serial uPCR measurements may be particularly useful in glomerular disease because of the higher variability of uACR and the greater cost of determining albumin in urine. Where appropriate, urine tests for Bence Jones protein (immunoglobulin light chains) may be required because this is not detected by standard proteinuria or albuminuria testing.

Kidney Imaging

Imaging of the kidneys with ultrasound is useful for a number of reasons. Small kidneys with reduced cortical thickness, showing increased echogenicity, scarring, or multiple cysts, suggest a chronic process. Structural abnormalities such as those observed in autosomal dominant polycystic kidney disease (ADPKD), hydronephrosis caused by obstruction, or coarse kidney scarring may be detected. NICE guidelines propose that kidney ultrasound scanning is important only in certain circumstances and suggest counseling patients if ADPKD is suspected before imaging.⁵ In some situations, imaging with computed tomography (CT), magnetic resonance (MR), or angiography may be useful, taking into account the risks of administering contrast media (see [Chapter 6](#)).

Further Investigations

Establishing the cause of CKD is important whenever possible, and further specific testing, as indicated by the history and results of initial investigations, may be required. There may be an underlying treatable condition that requires appropriate management, or there may be a genetic cause such as ADPKD, for which counseling should be offered. Furthermore, some kidney diseases may recur after transplantation (see [Chapter 113](#)) and an accurate diagnosis may therefore influence later management. Despite thorough investigation, however, the cause of CKD is often unclear, with an unhelpful medical history, minimal abnormalities on urinalysis, and small kidneys on ultrasound. In such patients, investigation should not be pursued relentlessly because the implications for treatment are often minimal. Attempting to obtain biopsy material from small kidneys is associated with risk, and even if a biopsy is performed, histologic assessment may simply show nonspecific chronic scarring rather than diagnostic features that explain the cause of kidney damage.

PREDICTING PROGNOSIS

With the cause of CKD established if possible, the GFR and the level of albuminuria measured, and other comorbidities categorized, it is possible to estimate the risk for CKD progression and likely future need for KRT. KDIGO recommends consideration of the GFR and the albuminuria categories together according to a heatmap of risk (see Fig. 82.2),² and there are online tools to assist with risk prediction using this approach (e.g., kidneyfailure.risk.com). Other factors associated with CKD progression may help inform prognosis. These include the cause of CKD, age, sex, ethnicity, dyslipidemia, smoking, obesity, history of CVD, ongoing exposure to nephrotoxic agents, and degree of control of hypertension and hyperglycemia. However, often the best guide to future change in kidney function is the previous pattern of decline, highlighting the importance of considering results of previous blood and urine testing during the initial assessment.

Monitoring and Defining Progression

Once CKD has been identified, arrangements should be made to ensure regular monitoring of kidney function and proteinuria. In patients at low risk for rapid eGFR decline, this can be done annually in either a primary or secondary care setting. However, assessment should be undertaken more regularly if the trajectory of the disease is not clear and in patients at higher risk for progression.

Determining a true change in kidney function over time may be difficult because small fluctuations in eGFR are common and not necessarily indicative of progression. These fluctuations may be caused by reversible factors, such as intravascular depletion or high meat intake, so repeat testing may be required. Both NICE and KDIGO guidelines define accelerated progression as a sustained decrease in GFR of 25% or more and a change in GFR category within 12 months.^{2,5} In addition, the NICE guidance refers to accelerated progression as a sustained decrease in GFR of 15 mL/min/1.73 m² or greater per year.⁵ In patients with accelerated CKD progression, consideration should be

given to detection of reversible causes (e.g., kidney tract obstruction) and a specialist referral may be required.

When to Refer to the Nephrologist

Chronic management of patients with early nonprogressive CKD is becoming the responsibility of primary care physicians in many well-developed health care systems, with follow-up in secondary care limited to those likely to progress to ESKD and require KRT. However, early assessment by nephrologists is useful for all patients newly diagnosed with CKD in whom a treatable underlying cause is suspected, even in those with advanced disease at presentation, to rule out treatable causes. Timely referral of those with progressive CKD allows preparation for dialysis, kidney transplantation, or initiation of a palliative approach if more appropriate. Substantially similar criteria for referral have been developed by NICE and KDIGO (Table 83.2). Such criteria are not absolute but should provide a guide to the primary care physician as to which patients are likely to benefit from specialist care. For example, many patients with stable category G4 CKD are successfully managed in the community, often after initial assessment by or with virtual advice from secondary care colleagues.

Unfortunately, a substantial proportion of patients with advanced CKD are referred late, often at the time when they need KRT. Late referral is often avoidable, although in some cases, patients may have had a truly silent illness or an acute presentation of a disease with rapid decline in kidney function.⁹ Over recent years, the introduction of routine reporting of eGFR in some health care systems has facilitated better communication between primary and secondary care providers and has led to a substantial fall in late referrals.¹⁰

Late presentation is disadvantageous to the patient because it limits the time to select the mode of dialysis or to be listed for “preemptive” kidney transplantation. There may be increased psychological stress, making it difficult for the patient to come to terms with the illness. Furthermore, because an arteriovenous fistula (AVF) takes several weeks to mature, patients presenting late start hemodialysis (HD) with

TABLE 83.2 Suggested Criteria for Referral of Patients With CKD to a Nephrologist

	NICE 2021	KDIGO 2012
Advanced CKD	Category G4 and G5 CKD A 5-year risk of needing kidney replacement therapy of >5% (measured using the 4-variable Kidney Failure Risk Equation)	Category G4 and G5 CKD
Proteinuria	High proteinuria: uACR ≥ 70 mg/mmol unless known to be caused by diabetes and appropriately treated	Consistent proteinuria: uACR ≥ 300 mg/g (≥ 30 mg/mmol)
Hematuria	Proteinuria (uACR ≥ 30 mg/mmol) together with hematuria	Urinary RBC casts, RBCs > 20/hpf sustained; not readily explained
Progression of CKD	Rapidly declining eGFR: Sustained decrease in GFR of ≥25% and a change in GFR category within 12 mo Sustained decrease in GFR of ≥15 mL/min/1.73 m ² within 12 mo NEW: 5-year risk of needing kidney replacement therapy of >5% (measured using the 4-variable Kidney Failure Risk Equation)	Progression of CKD: Sustained decrease in GFR of ≥25% and a change in GFR category within 12 mo Sustained decrease in GFR of ≥5 mL/min/1.73 m ² within 12 mo
Uncontrolled hypertension	Hypertension that remains poorly controlled despite the use of at least four antihypertensive drugs at therapeutic doses	CKD and hypertension refractory to treatment with four or more antihypertensive agents
Hereditary kidney disease	Known or suspected rare or genetic causes of CKD	Hereditary kidney disease
Other conditions	Suspected renal artery stenosis	Recurrent or extensive nephrolithiasis Persistent abnormalities of serum potassium

CKD, Chronic kidney disease; eGFR, estimated glomerular filtration rate; hpf, high power field; KDIGO, Kidney Disease: Improving Global Outcomes; NICE, National Institute for Health and Care Excellence; RBC, red blood cell; uACR, urinary albumin-to-creatinine ratio.

Data from Kidney Disease: Improving Global Outcomes [KDIGO] CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150; and National Institute for Health and Care Excellence. Chronic kidney disease assessment and management. 2021. NICE guideline [NG203].

central venous catheters. Catheters are prone to infectious complications and inevitably damage central veins, leading to thromboses and stenoses, which may manifest at a later stage when venous return from one or the other arm is increased by the subsequent construction of an AVF (see [Chapter 96](#)).¹¹ Late presentation of CKD also precludes effective treatment of complications such as hypertension and anemia, which may contribute to CVD and ultimately limit life span.¹² Most importantly, late referral is associated with greater subsequent costs of medical care and a worse prognosis.¹³

PREVENTION OF CHRONIC KIDNEY DISEASE PROGRESSION

Management of CKD should be aimed at slowing the rate of decline of kidney function and minimizing the effects of other complications. Except for specific management of the underlying kidney disease where possible, the most effective intervention is control of blood pressure (BP) and introduction of nephroprotective medications including (in patients with albuminuria) angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs; see [Chapter 82](#)) and sodium glucose cotransporter 2 (SGLT2) inhibitors. Control of glycemia in patients with diabetes and CKD is covered in [Chapter 33](#).

Hypertension and Renin Angiotensin Aldosterone System Blockade

Hypertension is very common in patients with CKD, the level of BP being associated with the rate of loss of kidney function,¹⁴ and BP control slows the rate of decline (see [Chapter 82](#)). BP targets have changed to reflect the available evidence. Although observational studies link low BP to higher mortality,¹⁵ interventional trials such as SPRINT have demonstrated the advantages of more intensive BP control in CKD patients.¹⁶ Guidelines by both NICE and KDIGO emphasize the importance of considering coexistent CVD, other comorbidities, and side effects when choosing medications and BP targets, particularly for elderly patients.^{5,17} Lifestyle modifications should be encouraged, including maintenance of a healthy weight, reductions in salt and alcohol intake, and regular exercise (see [Chapter 36](#)).

Although current recommendations are based on office BP recordings, these should be standardized if possible (i.e., obtained according to a recommended preparation protocol).¹⁸ Home and ambulatory monitoring readings, where available, should be used to complement standardized office recordings. It has been shown that up to 30% of CKD patients who were thought to have hypertension have normal BPs at home, whereas 40% of patients who were thought to be normotensive (or to have adequately treated hypertension) were hypertensive at home.¹⁹ Although ambulatory monitoring is not universally recommended, there should be a low threshold for undertaking 24-hour monitoring or asking patients to undertake self-measurements at home if they prefer.

Target BP levels and antihypertensive therapy in CKD patients are discussed in [Chapter 82](#). ACE inhibitors or ARBs are recommended as first-line agents for patients with evidence of proteinuria, both to slow progression of CKD and to reduce CV risk.²⁰ However, multidrug regimens are usually required to obtain good control. Patients with CKD are vulnerable to drug side effects, particularly during intercurrent illness, when they may develop hyperkalemia and AKI. The KDIGO guidelines recommend the temporary discontinuation of certain drugs that may exacerbate AKI or are prone to increased adverse events at low GFR (including ACE inhibitors, ARBs, aldosterone inhibitors, direct renin inhibitors, diuretics, nonsteroidal anti-inflammatory drugs, metformin, lithium, and digoxin) in patients with a GFR of less than 60 mL/min/1.73 m² (CKD categories G3a–G5) who have serious intercurrent illness.

Sodium Glucose Cotransporter 2 Inhibitors and Mineralocorticoid Receptor Antagonists

SGLT2 inhibitors enhance urinary excretion of sodium and glucose by blocking their uptake in the proximal tubule. Although originally developed as drugs to improve glycemic control in patients with type 2 diabetes (see [Chapter 33](#)), these agents have both kidney and cardioprotective effects that extend to patients with CKD who do not have diabetes.²¹ A new nonsteroidal mineralocorticoid receptor antagonist (Finerenone) has also recently been demonstrated to have cardiorenal benefits in patients with CKD and type 2 diabetes.²² It seems likely that these newer agents will be used more extensively in the future in the management of CKD in addition to ACE inhibitors and ARBs.

Dietary Advice

Detailed dietary advice and education, along with ongoing support from an appropriately trained professional, are important in the management of patients with CKD. Obesity is associated with a more rapid decline of kidney function, so in early CKD, weight loss may be appropriate. However, in advanced CKD, malnutrition is common (see [Chapter 90](#)). The causes are multifactorial but include anorexia, acidosis, insulin resistance, inflammation, oxidative stress, and urinary protein loss. Biochemical indicators may demonstrate a decrease in serum albumin, transferrin, and cholesterol. Weight should be monitored in patients who progress to CKD categories G4 and G5. Serum creatinine concentrations, which, in part, reflect muscle mass, may stop rising despite a progressive loss of kidney function because of compromised nutritional status.

In light of this, recommendations to restrict protein intake are controversial. Although reduced protein intake may slow progression of decline of kidney function, many patients develop protein-calorie malnutrition on a low-protein diet.²³ KDIGO has recommended that protein intake should be lowered to 0.8 g/kg/day in adults with CKD and GFR less than 30 mL/min/1.73 m², whereas high protein intake (>1.3 g/kg/day) should be avoided in adults with CKD at risk for progression. When this recommendation is followed, detailed dietary assessment and supervision are needed to ensure that malnutrition is prevented (see [Chapter 90](#)).

One of the earliest effects of CKD is to limit the ability of the kidney to compensate for large changes in sodium and water intake (see [Chapter 8](#)). Salt and water retention are major factors contributing to hypertension in CKD patients and, in more advanced stages, to morbidity and mortality through systemic or pulmonary edema. Therefore, sodium intake should ideally be restricted to less than 90 mmol/day (5 g/day of sodium chloride), except in salt-wasting conditions. Advice about optimal fluid intake at each stage of CKD is needed to prevent volume overload. Salt substitutes containing potassium should be avoided because of the risk for hyperkalemia. In stages G4 and G5 CKD, education and advice about restriction of potassium and phosphate may be required.

MANAGEMENT OF COMPLICATIONS OF CHRONIC KIDNEY DISEASE

A detailed discussion of the complications of CKD is provided in [Chapters 84 to 92](#). With the exception of hypertension, there are usually few clinical manifestations associated with CKD stages G1 and G2 (GFR > 60 mL/min/1.73 m²). Other complications (discussed in the following sections) tend to develop progressively as GFR declines to less than 60 and in particular less than 30 mL/min/1.73 m² (i.e., during CKD stages G4 and G5).

Anemia

Anemia is common in CKD stages G3a to G5 and is caused by a relative deficiency of erythropoietin, although reduced availability of iron and chronic inflammation are frequent contributory factors (see [Chapter 86](#)). Anemia may have multiple adverse effects, including worsening cardiac dysfunction by increasing cardiac output and exacerbating left ventricular hypertrophy, thereby accelerating the decline of kidney function. Anemia may also reduce cognition and concentration. However, clear evidence that reversal of anemia using erythropoiesis-stimulating agents (ESAs) is associated with improved clinical outcomes is lacking, and randomized trials have suggested that in some circumstances these agents may cause harm.²⁴ A new approach to the treatment of renal anemia is the use of oral small-molecule stabilizers of hypoxia inducible factor. These agents, which are currently being evaluated in phase 3 studies, could offer a simpler and cheaper therapy compared with ESAs, with potential benefits beyond increases in hemoglobin (Hb) level.²⁵

The relevant KDIGO guideline recommends that all patients identified as having CKD categories G3a and below should have their Hb levels monitored annually, increasing to twice a year for stages G4 and G5.²⁶ Anemia in adults is diagnosed when the Hb concentration is less than 13.0 g/dL in males and less than 12.0 g/dL in females. NICE recommends that management of anemia should be considered in patients with CKD when the Hb level is up to 11 g/dL.⁵ In anemic patients, investigations for other causes should be conducted, including measurement of iron stores, serum vitamin B₁₂, and folate levels. ESAs should not be started until treatment of iron deficiency or other underlying causes have been addressed and then only after considering the balance of benefits (from the reduced requirement for blood transfusions and abrogation of anemia-related symptoms) against the potential harms, which may include an increased risk for stroke and malignancy.

If anemia does not respond to correction of underlying causes, such as iron deficiency, KDIGO recommends that ESAs be commenced when Hb concentrations are less than 10.0 g/dL, if indicated.²⁶ Hb target ranges are discussed in [Chapter 86](#).

Bone and Mineral Metabolism

Hyperphosphatemia, together with a deficiency of 1,25-dihydroxyvitamin D₃, can contribute to secondary hyperparathyroidism and ultimately to the development of renal bone disease. These biochemical and endocrine changes, in association with the closely related histologic abnormalities of bone and soft tissue calcification, are collectively termed the *CKD–mineral and bone disorder* (see [Chapter 88](#)).²⁷ Bone disease may already be manifested in CKD category G3b and may be well established in ESKD, even though patients may remain asymptomatic. In addition to the need to prevent bone complications, active management of CKD–mineral and bone disorder may help prevent some of the CV complications of CKD.²⁸

KDIGO recommends measuring serum levels of calcium, phosphate, PTH, and alkaline phosphatase activity in adults with GFR less than 45 mL/min/1.73 m² (GFR stages G3b–G5). Determining the optimal level of PTH in CKD has been controversial. For patients with levels of intact PTH above the upper normal limit of the assay, efforts should be made to correct hyperphosphatemia, hypocalcemia, and vitamin D deficiency if present. KDIGO guidelines recommend that serum phosphate concentrations be maintained in the normal range according to local laboratory reference values,²⁹ whereas the UK Kidney Association support a level between 0.9 and 1.5 mmol/L for patients with category G4 and G5 CKD.³⁰ Early advice on dietary phosphate management by a specialist dietician or other professional is important in helping patients achieve this. Phosphate-binding drugs may be required,

and their choice is discussed in [Chapter 88](#). Prescription of vitamin D supplements or analogs, in the absence of documented deficiency, to suppress elevated PTH concentrations in patients with CKD not on dialysis is not recommended.²⁹ Calcimimetics are a group of drugs that mimic the action of calcium on parathyroid glands by activation of the calcium-sensing receptor and thus reduce the release of PTH. At present, cinacalcet (an oral calcimimetic) and etelcalcetide (which is given intravenously) are generally used in patients receiving dialysis, particularly those that are unfit for surgical parathyroidectomy. Whether these drugs have additional health benefits is not clear (see [Chapter 88](#)).

Metabolic Acidosis

The metabolic acidosis associated with CKD is caused by failure of hydrogen ion excretion and may be compounded by the accumulation of organic acids and bicarbonate loss, particularly in interstitial kidney diseases. Clinical symptoms resulting from acidosis are rare until patients reach CKD stage G5, when dyspnea may occur as a result of respiratory compensation. Other causes of dyspnea in advanced CKD, such as anemia and pulmonary edema, should always be considered. Acidosis aggravates hyperkalemia, inhibits protein anabolism, and accelerates calcium loss from bone where the hydrogen ions are buffered.³¹ Correction of metabolic acidosis may slow progression of kidney disease,³² although larger trials are required to confirm this.

KDIGO recommends that in patients with CKD and serum bicarbonate concentrations less than 22 mmol/L (NICE recommends a threshold of <20 mmol/L), oral bicarbonate supplementation should be given to maintain serum bicarbonate within the normal range, unless contraindicated. However, the associated sodium loading may aggravate hypertension and fluid retention, and severe metabolic acidosis associated with symptoms in a patient with CKD stage G5 may be an indication to start dialysis.² Novel pipeline drugs include an oral polymer designed to remove acid from the body with high capacity and specificity. This raises the possibility of treatment of acidosis without the problems associated with increased sodium intake.

Cardiovascular Risk

Patients with CKD have an increased prevalence of CVD and are far more likely to die of a CVD-related cause than to progress to ESKD (see [Chapter 85](#)). Therefore, appropriate management of existing CVD and minimization of future CV risk is vital for all patients with CKD. Agents used to slow progression of CKD, including ACE/ARBs, SGLT2 inhibitors, and mineralocorticoid receptor antagonists (MRAs), may themselves have cardiovascular benefits in the context of CKD.

Unfortunately, many trials of interventions for CVD have excluded patients with CKD,³³ and there is doubt about the relevance of existing standards of care of CVD to patients with CKD.³⁴ Nonetheless, the level of care for coronary heart disease offered to patients with CKD should not be prejudiced by their CKD. NICE suggests that antiplatelet drugs should be offered to patients with CKD for the secondary prevention of CVD, and some experts would extend this recommendation to primary prevention for those at risk for atherosclerotic events. However, there is an increased risk for minor bleeding, and a recent systematic review of antiplatelet agents for patients with CKD found that although the incidence of myocardial infarction is reduced, major bleeding is increased.³⁵ Thus, the risks may outweigh benefits among individuals with low risk for CV events, including those with early stages of CKD who do not have clinically evident occlusive arterial disease.

For reduction of cardiovascular risk, KDIGO guidelines recommend that statin treatment be routinely offered to patients with CKD who are aged older than 50 years and to younger patients with

additional risk factors, irrespective of baseline lipid values.³⁶ NICE recommends that all people with CKD should be offered atorvastatin 20 mg for the primary or secondary prevention of CVD.³⁷

Risk for Infections

Infection is the second most common cause of death after CVD in patients with ESKD. This is, in part, because of defects in both cellular and humoral immunity, which make CKD a state of chronic immunosuppression (see [Chapter 87](#)).³⁸ T-cell responses to de novo antigens are deficient, partly because of impaired antigen presentation by monocytes. Neutrophil activation is defective, and although serum immunoglobulin levels are usually normal, antibody responses to immunization may be poor. Patients with CKD have an increased susceptibility to bacterial infection (particularly staphylococcal), increased risk for reactivation of tuberculosis (typically with a negative tuberculin skin test response), and failure to eliminate hepatitis B and C viruses after infection.

In view of these increased risks, the KDIGO guidelines recommend that all adults with CKD be offered annual vaccination with influenza vaccine unless contraindicated and that all adults with eGFR less than 30 mL/min/1.73 m² (GFR categories G4–G5) and those at high risk for pneumococcal infection (e.g., patients with nephrotic syndrome, with diabetes, or who are receiving immunosuppression) receive vaccination with polyvalent pneumococcal vaccine unless contraindicated.² Patients who have received pneumococcal vaccination are offered revaccination within 5 years. In addition, those at high risk for progression of CKD with eGFR less than 30 mL/min/1.73 m² (GFR categories G4–G5) should be immunized against hepatitis B and the response confirmed by appropriate serologic testing. This should happen as early as possible to maximize the chances of seroconversion.³⁹ Response to COVID-19 vaccination is also attenuated in CKD.⁴⁰

Gastrointestinal Problems in Chronic Kidney Disease

Gastrointestinal (GI) symptoms and disease are common in patients with CKD ([Table 83.3](#)). Anorexia, nausea, and vomiting related to advanced CKD may contribute to malnutrition and wasting, which, in turn, herald an adverse prognosis (see [Chapter 90](#)). In addition, uremia-associated reduced taste sensation or abnormal taste can impair dietary intake. Gastroesophageal reflux is also common in patients with CKD and related to GI tract dysmotility or delayed gastric emptying, in particular in diabetes or amyloidosis or patients on peritoneal dialysis, given their increased intra-abdominal pressure.

Extensive peptic ulcer disease, gastritis, and duodenitis occur more commonly in CKD than in the general population, with the highest incidence in dialysis patients.⁴¹ Peptic ulcers are often multilobar and postbulbar, but pain is less frequent. *Helicobacter pylori* infection is not increased in CKD. There is a risk for excess calcium and magnesium absorption with some antacids in CKD, and aluminum- or bismuth-containing preparations should be avoided, given their potential for accumulation in CKD.

In the large bowel, the incidence of diverticular disease is increased in patients with ADPKD (see [Chapter 46](#)). Constipation is common in CKD and results from dietary restrictions, restricted fluid intake, electrolyte abnormalities (including hypercalcemia), and commonly prescribed drugs, such as calcium-based phosphate binders, sevelamer, oral iron, opioid analgesics, and calcium resonium. In extreme cases, constipation may result in intestinal pseudo-obstruction, which may also be acutely precipitated by surgery and retroperitoneal hemorrhage. Patients with CKD are at risk for more frequent or severe *Clostridium difficile* infection, slower treatment response, and more frequent relapse and have a higher risk for death.⁴²

Intestinal ischemia is an important cause of an acute abdomen in older CKD patients. Some cases result from nonocclusive mesenteric

TABLE 83.3 Important Causes of Common Gastrointestinal Symptoms in Patients With CKD

Clinical Feature	Causes
Anorexia	Uremic toxicity Inadequate dialysis clearances Delayed gastric emptying
Nausea and vomiting	Uremic toxicity Delayed gastric emptying/gastroparesis Gastritis, duodenitis Peptic ulcer disease Drugs
Constipation	Drugs, including opioid analgesia GI pseudo-obstruction Diverticular disease
Diarrhea	Diabetic enteropathy Diverticular disease <i>Clostridium difficile</i> infection Dialysis-related amyloidosis (very rare)
GI hemorrhage	Gastritis, duodenitis Esophagitis Peptic ulcer disease Angiodysplasia Intestinal ischemia Vasculitis Dialysis-related amyloidosis (very rare)
Acute abdominal pain	Gastritis, duodenitis Complications of peptic ulcer disease Acute pancreatitis Intestinal ischemia Diverticulitis GI pseudo-obstruction Colonic perforation from fecal impaction Complications of peritoneal dialysis (peritonitis, dialysis catheter malposition, dialysate infusion or drain pain) Complications of autosomal dominant polycystic kidney disease Retroperitoneal hemorrhage

CKD, Chronic kidney disease; GI, gastrointestinal.

ischemia. Predisposing factors include hypotension, cardiac failure, hypoxia, increased plasma viscosity, and constipation. Abdominal examination can be misleadingly benign at presentation, but often peripheral neutrophil leukocytosis and progressive lactic acidosis are present. GI hemorrhage resulting from peptic ulcers, gastritis, duodenitis, angiodysplasia ([Fig. 83.2](#)), or nonulcer, nonvariceal bleeding because of uremic hemostatic defects is also common in CKD and is associated with high mortality. Idiopathic ascites with a high protein content may occur in HD patients, sometimes related to inadequate dialysis dose. The diagnosis is by exclusion of other causes. Management includes optimization of volume and small-solute clearance. The condition may resolve after transplantation or a modality switch to PD.

CARE OF THE PATIENT WITH PROGRESSIVE CHRONIC KIDNEY DISEASE

To optimize the care of patients with progressive CKD, management is provided in a multidisciplinary setting where a range of professionals are able to provide education and information about diet, different

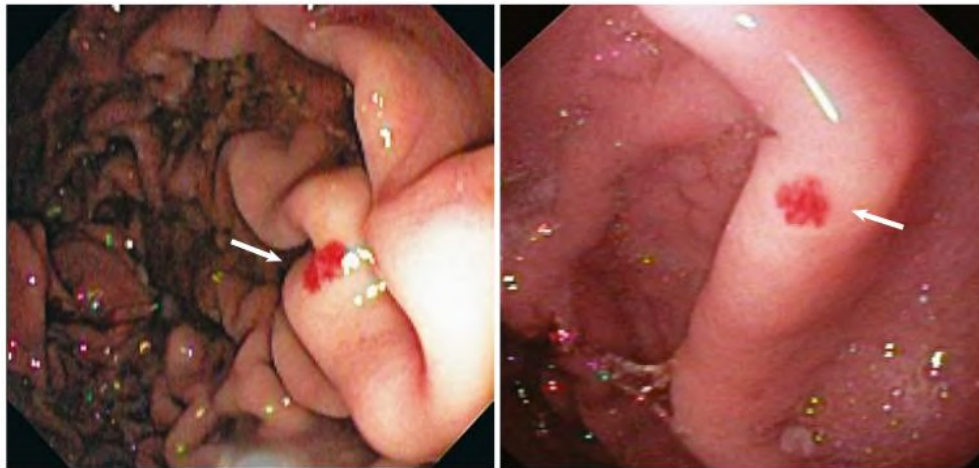


Fig. 83.2 Examples of gastric angiodysplasia (arrows) in a dialysis patient. (Courtesy Drs. R. Winograd and C. Trautwein, Aachen, Germany.)

KRT modalities, transplant options, vascular access surgery, and social care. Psychological comorbidity is common among patients with CKD. Health care professionals working with patients with CKD should take account of the psychological aspects of coping with the condition and offer access to support groups, counseling, or a specialist nurse. The aim is to create an environment in which patients can become informed and proactive in their care.

Chronic Kidney Disease and Risk of Acute Kidney Injury

All patients with CKD are at increased risk for AKI, and AKI is associated with the development and progression of CKD.^{43,44} Imaging studies that require iodinated radiocontrast media carry a risk for AKI, and the benefit of a diagnostic scan needs to be balanced against the risks (Chapter 6). If the investigation is needed, the lowest dose of radiocontrast should be used, the patient should be adequately hydrated, and potentially nephrotoxic agents should be withdrawn before and after the procedure; however, the fear of radiocontrast-induced AKI should not prevent or impair a necessary diagnostic workup. The potential risk for nephrogenic systemic fibrosis from gadolinium-based contrast media and measures to reduce it are also discussed in Chapter 6.

Many commonly used medications increase the risk for AKI, and the level of GFR should be considered when any drug is prescribed. As discussed earlier, the need for temporary cessation of some medications should be considered during periods of severe intercurrent illness. Other causes of reduction in kidney perfusion can lead to AKI, including volume depletion from excessive diuretics, insufficient fluid intake in hot weather, diarrhea or vomiting, heart failure, myocardial infarction, and tachyarrhythmias. Severe hypercalcemia, resulting from either coadministration of high doses of vitamin D and calcium-containing phosphate binders or from underlying disease, can also cause AKI.

Clinicians should always consider whether acceleration of loss of kidney function is a result of relapse of the underlying disease or of a superimposed problem such as acute interstitial nephritis (see Chapter 64), obstructive uropathy (see Chapter 61), or renal vein thrombosis.

Timing the Initiation of Kidney Replacement Therapy

Despite all attempts to optimize the management of CKD, many patients will progress to needing KRT. Patients with eGFR less than 20 mL/min/1.73 m² and/or who are likely to progress to ESKD within 12 months should receive education and counseling, with the support of a multidisciplinary team, to aid their selection of the most appropriate

KRT modality (see Chapter 95). If HD is the preferred option, an AVF should be constructed, remembering that it may take 8 to 12 weeks for veins to become adequately arterialized before needling can be attempted (see Chapter 96). Similar plans need to be made for pre-emptive insertion of a peritoneal dialysis catheter to allow time for healing and training before any acute need for commencement of dialysis (see Chapter 101).

Early kidney transplantation may be associated with improved long-term outcome,⁴⁵ so patients should be assessed for their suitability and, when feasible, activated on the waiting list before dialysis is commenced. This maximizes the chances of the potential recipient remaining in reasonable health. The availability of a living donor should be explored to increase the chances of preemptive transplantation before the patient begins dialysis. The KDIGO guidelines recommend that living donor preemptive kidney transplantation in adults be considered when GFR is less than 20 mL/min/1.73 m² and there is evidence of progressive and irreversible CKD over the preceding 6 to 12 months.²

Planned early initiation of dialysis is not associated with improvement in outcomes compared with commencement when indicated by signs and symptoms of uremia.⁴⁶ KDIGO suggests that dialysis be initiated when one or more of the following are present: symptoms or signs attributable to kidney failure (serositis, acid-base or electrolyte abnormalities, pruritus); inability to control volume status or BP; a progressive deterioration in nutritional status refractory to dietary intervention; or cognitive impairment.² These problems often but not invariably occur when the GFR is less than 15 mL/min/1.73 m² (see Chapter 95).

Conservative Management

The potential burden of commencing KRT in terms of high short-term mortality rates, recurrent hospitalizations, time spent traveling, and limited improvement in quality of life for some elderly patients and those with multiple comorbid disease is increasingly recognized. This has led to the practice of offering patients with incipient kidney failure the option of choosing not to start dialysis but to maintain ongoing follow-up and symptomatic support through conservative management. Although dialysis may offer longer survival, those choosing conservative management may have as many hospital-free days as those who choose HD.⁴⁷ The symptoms of advanced uremia can be distressing, and it is important to ensure that patients who choose this pathway have access to members of the multidisciplinary team with expertise in palliative care to facilitate a death free of suffering.

SELF-ASSESSMENT QUESTIONS

1. Which of the following is the most common cause of death for people with CKD and $\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$?
 - A. Failure of dialysis access
 - B. Sepsis
 - C. Withdrawal from dialysis
 - D. Cardiovascular disease
 - E. Cerebrovascular disease
2. The equation recommended by KDIGO for estimating GFR is:
 - A. The Modification of Diet in Renal Disease (MDRD) equation
 - B. The Cockcroft-Gault formula
 - C. The CKD-EPI formula
 - D. The Mayo Clinic Quadratic formula
 - E. The Schwartz formula
3. Which of the following variables is *not* needed to calculate eGFR from the CKD-EPI formula?
 - A. Age
 - B. Weight
 - C. Race
 - D. Sex
 - E. Creatinine
4. Which of the following symptoms is suggestive of CKD, regardless of cause?
 - A. Pain in the renal angle
 - B. Nocturia
 - C. Urinary frequency
 - D. Tremor
 - E. Anuria
5. In a well patient, which of the following biochemical or hematologic abnormalities is suggestive of CKD rather than acute kidney injury?
 - A. Calcium 3.5 mmol/L
 - B. Potassium 7.5 mmol/L
 - C. Hemoglobin 10.7 g/dL
 - D. Creatinine 13.5 mg/dL
 - E. Sodium 132 mmol/L

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